

Cloud - Cohort Ascend Week1 - 8 February 2026

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Week1 - Introduction to Big Data

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The data which satisfy all these 5 Vs.

1) Volume - Huge Amount of data

Example - Peta Bytes in social media, or ecommerce platforms

2) Variety :

Structured Form : RDBMS Table : MySQL Database

Semi Structured : Excel File , JSON , XML

Unstructured: Log Files : Images , Videos

3) Velocity : Speed at which data is coming in

Ex : Amazon sale how many phones were sold in 1 Hour

4) Veracity : Quality and correctness of data

Eg : if age is coming is as null and data is messy which we have to correct applying some rules

5) Value : The data should be able to generate some value in help in decision making.

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Monolithic Vs Distributed System

1) Mono means (Vertical Scaling)

One Big System Holding a lot of power

Resources :

Computer : CPU (4 Core)

Memory (Ram - 8GB)

Storage : hard Disk 1TB

X Resources - Y Performance

2X Resources < 2Y Performance

Monolithic Is not a scalable system. There is limited to extend in resources

2) Distributed Systems (Horizontal Scaling)

A group of machines or a 5 node system

In which we can add more resources

Example adding 6th node to a cluster

Here x Resource - Y performance
2X resource - 2Y Performance

All big data systems are based on this architecture as it is scalable and good big data system depend on below points

- 1) Storage : where I am going to store the data
- 2) Processing : How I am going to process the data
- 3) Scalability : is my system scalable if increasing traffic

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Hadoop

Is a framework which is used to solve big data problems

2007 : Hadoop 1.0

2012 : Hadoop 2.0

Currently Hadoop 3

- 1) Storage - HDFS - Hadoop Distributed File System
- 2) MapReduce : Processing of Big Data in Java Language
- 3) Yarn : Yet another resource manager (Resource Manager)

1) Sqoop is for data movement

Example - if you have some data in HDFS as a table and you want to transfer it to MySQL database.

2) PIG - is scripting language and mainly used to clean data

3) Hive : Provide a SQL Type interface to query the data

4) Oozie : Workflow Scheduler

Like we have a 2 map reduce jobs and they have to be executed at 8PM after coming of data
The automation will be done by the oozie

5) Hbase : is NOSQL Database in Hadoop .

Challenges with Hadoop

1) Map Reduce is very slow and it is very hard to write JAVA Code.

2) We need to learn so many different components and each components has its own big learning curve.

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Cloud and its advantages

50 Servers

You need a space to keep these servers

You buy 50 nodes

Cooling Equipment

Maintenance and Admin Team to check

Costly

Cloud Providers

AWS - 70% Hold Portion

Azure

GCP

Cloud - Take it like infrastructure on rent

Advantages :

- 1) Scalable : we can expand the machines in the cluster
- 2) Capital Vs operational expenditure : you spend small amount every month and do not pay infra installation and operational cost is given.
- 3) Agility :
- 4) Geo Distribution : 1 Server in US and 1 server in India.
It will meet the requirement in case of natural calamity or disaster.
- 5) Cost effective

Types Of cloud :

- 1) Public ; AWS Azure , GCP

All have their public offering and services

- 2) Private : any organization who thought their data is confidential or do not want to move to public cloud. They use private cloud .
- 3) On premise : when you buy the infrastructure and everything is all yours
VMS, Machines , Engineers
- 4) Hybrid Cloud - Sensitive data in private, Less sensitive in public

